MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
SECOND SEMESTER PRACTICAL EXAMINATION
(2024 ADMISSION ONWARDS)

MG2DSCCSC100- PYTHON PROGRAMMING

Duration of the Examination: 1.5 Hrs Maximum Marks: 35

List of Questions

1. Write a Python Program to prompt for a score between 0.0 and 1.0. If the score is out of range, print an error. If the score is between 0.0 and 1.0, print a grade using the following table.

Score	Grade
>= 0.9	A
>= 0.8	B
>= 0.7	C
>= 0.6	D
< 0.6	F

[A][CO4]

2. Write a Python Program to check if a given year is a Leap Year. [A][CO4]
3. Write a Python program to check whether a number is palindrome or not. [A][CO4]
4. Write a Python Program to find the sum of digits in a number. [A][CO4]
5. Write a Python Program to display the Fibonacci Sequences up to n^{th} term where n is provided by the user. [A][CO4]
6. Write a Python Program to find the sum of all odd and even numbers up to a number specified by the user. [A][CO4]
7. Write a Python Program to find the factorial of a number. [A][CO4]
8. Write a Python Program to check whether a number is Prime or not. [A][CO4]
9. Write a Python Program to find the area of Trapezium using the formula Area = (1/2) * (a + b) * h,where a and b are the 2 bases of Trapezium and h is the height (Use function) [A][CO4]
10. Write a Python Program to check if a 3 digit number is Armstrong number or not using functions. [A][CO4]
11. Write a Python program to find the largest of three numbers using functions. [A][CO4]
12. Write a Python program using functions to find the value of ${}^{n}P_{r}$ and ${}^{n}C_{r}$. [A][CO4]
13. Write Python Program to sort numbers in a List in ascending order. [A][CO4]
14. Write Python Program to search for a number in the List and report 'Success' or 'Failure'. [A][CO4]

15. Write a python Program to find Mean, Variance and Standard Deviation of numbers in a List. [A][CO4]
16. Write a Python Program to add two matrices using List. [A][CO4]
17. Create a list containing three elements, and then create a tuple from that list. [A][CO4]
18. Write a Python program to store the latitude and longitude of a place as a tuple and display it. [A][CO4]
19. Write a program that takes a range and creates a list of tuples within that range with the first element as the number and the second element as the square of the number. [A][CO4]
20. A school maintains a tuple of student records, where each record consists of a student's name, age, and marks in three subjects. Write a Python program to display the details of all students, Search for a particular student and Calculate and print the average marks of each student. [A][CO4]
21. Write a Python Program to generate a Dictionary that contains (i: i*i) such that i is a number ranging from 1 to n. [A][CO4]
22. Write a Python Program that accepts a sentence and calculate the number of digits, uppercase and lowercase letters(Use Dictionary). [A][CO4]
23. Write a Python program to input information about a few employees as given below:
    a. Name
    b. Employee Id
    c. Salary

    The program should output the employee ID and salary of a specified employee, given his name (Use Dictionary). [A][CO4]
24. Write Python code to create a dictionary that accepts a country name as a key and its capital city as the value. Display the details in sorted order. [A][CO4]
25. Write a program that has the dictionary of student names as keys and phone numbers as its values. Print the dictionary in a sorted order. Prompt the user to enter the name and check if it is present in the dictionary. If the name is not present, then enter the details in the dictionary. [A][CO4]
26. Write a Python program to find the length of the set. [A][CO4]
27. Write a program to create an intersection, union, set difference, and symmetricdifference of sets.
    [A][CO4]
28. Write a program to read the first n lines of a file. Prompt the user to enter the value for n. [A][CO4]
29. Write a program that reads the contents of the file and counts the occurrences of each letter. Prompt the user to enter the filename. [A][CO4]
30. Write a program to read and write the contents from one csv file to another. [A][CO4]